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RESEARCH ARTICLE

DIR-YOLO: Segmentation of Alveolar Bone and Mandibular Canal in CBCT Images for Dental Implant Recommendation

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Health Research Ethics Committee of Universitas Airlangga Academic Dental Hospital under Application No. 00450/HRECC.FODM/IV/2024, and performed in line with the Seven ethical standards of the World Health Organization (WHO).

ABSTRACT In dental implant planning, segmentation of the alveolar bone (AB) and mandibular canal (MC) is essential to identify a safe area for dental implant placement in the edentulous molar region. This anatomical information serves as the basis for radiologists to determine the most appropriate dental implant dimension, including its diameter and height. Cone-Beam Computed Tomography (CBCT) is a 3D imaging modality that provides high-resolution cross-sectional images of the jaw. It enables detailed visualization of AB and MC. The dental radiologists manually identify the AB and MC to determine the suitable dental implant dimension. This process is time-consuming and its accuracy depends on the dental radiologist's skill and experience. This study proposes an automated method for segmenting AB and MC from CBCT images to support dental implant dimension recommendation for molar teeth. This study introduces DIR-YOLO (Dental Implant Recommendation-YOLO), an efficient AB and MC segmentation model based on a modified YOLOv8 architecture. The proposed method enhances YOLOv8 by replacing the original C2f module with C3Ghost, replacing the standard convolution with Ghost Convolution to reduce computational complexity, and simplifying the segmentation head to use only two-scale feature maps. It focuses on low and high resolutions for optimized AB and MC segmentation performance. DIR-YOLO achieved a mean Dice Similarity Coefficient (mDSC) of 92.12% and mean Hausdorff Distance (mHD) of 0.96 mm, indicating high AB and MC segmentation performance. Furthermore, Weighted Cohen's Kappa and the Kruskal-Wallis test demonstrate a high level of statistical agreement between the proposed method and the dental radiologists and indicating no significant differences in the recommended dental implant dimensions. These results indicate that DIR-YOLO can assist dental radiologists by automating AB and MC segmentation for supporting accurate dental implant dimension recommendation for molar teeth.

INDEX TERMS Alveolar bone, mandibular canal, dental implant recommendation, CBCT, segmentation, YOLO.

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I. INTRODUCTION

A dental implant is an artificial tooth root placed in the jaw to support a replacement tooth. Dental implant planning involves identifying the optimal dental implant placement

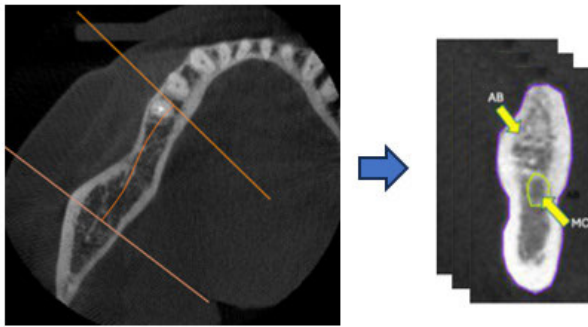


FIGURE 1. Cross-sectional CBCT images showing AB and MC areas in the edentulous region.

area and determining the appropriate dental implant dimension that ensure both functionality and safety. Accurate planning not only improves the success rate of dental implants but also reduces the risk of complications such as nerve injury or implant failure [1].

Cone-Beam Computed Tomography (CBCT) is widely used in dental implant planning as the standard imaging technique. It produces high-resolution 3D cross-sectional images of the jaw [2], [3]. It visualizes important anatomical structures such as the Alveolar Bone (AB) and Mandibular Canal (MC) [4]. AB refers to the part of the jawbone that contains the tooth sockets. It provides support for teeth and implants [5]. MC houses the inferior alveolar nerve and blood vessels, running within the lower jawbone [6], [7]. These two areas are important when placing dental implants, particularly in edentulous molar regions (area where molar tooth are missing) [8]. Fig. 1 shows the AB and MC structure from cross-sectional CBCT images taken in the edentulous molar region. These cross-sectional CBCT images are generated by slicing perpendicular to the jaw curve. In the edentulous region, multiple slices are produced, each providing a different anatomical view along the missing tooth area. Accurate identification of AB and MC in these slices is essential for determining the appropriate dental implant dimensions, including both diameter and height.

In clinical practice, dental implant planning for molar teeth by a dental radiologist begins with identifying the AB and MC using cross-sectional CBCT images [9]. This step is especially important in edentulous molar regions, where molar teeth are missing and implants need to be placed carefully. The dental implant must be positioned within the AB and at a safe distance from the top of the MC to avoid nerve injury [10]. After identifying the AB and MC, the dental radiologist selects the most suitable dental implant with optimal diameter and height. This process is performed manually, which requires time and relies on the radiologist's expertise and experience [11].

Recently, deep learning techniques have been increasingly applied in medical image processing [12], dental imaging, and implant planning tasks [13]. Previous studies have applied deep learning methods to support dental implant

planning by automatically identifying key anatomical structures such as AB, MC, and edentulous regions in CBCT modalities. Cui et al. [14] developed a fully automatic AI system for segmenting teeth and AB from 3D CBCT images using a multi-stage pipeline based on V-Net architectures. Their method achieved high accuracy, with Dice Similarity Coefficient (DSC) of 94.1% for teeth and 94.5% for alveolar bone. Gan et al. [15] proposed a two-step method to segment tooth and AB from dental CT images. The approach first segments the combined bony tissue using a global convex level set model then separates individual tooth and AB using a Radon transform and local level set. The method achieved high segmentation accuracy with a DSC of 95.64% for AB. This shows the method's potential for accurate 3D model reconstruction in orthodontic planning.

Jaskari et al. [16] used a 3D fully convolutional network to segment the MC from CBCT images, achieving dice DSC of 57%. Several studies have also applied deep learning methods for MC segmentation on 2D CBCT slices. Liu et al. [17] used U-Net and achieved a DSC of 92.48% and IoU of 90.03%. Wang and Feng [18] applied U-Net and obtained a DSC of 81.88%. Faradhilla et al. [19] proposed Residual Fully Convolutional Network (RFCN) with dual attention, reaching a DSC of 89.1%. Chun et al. [20] implemented DenseNet121 as the U-Net backbone, achieving a DSC of 86.1%, an IoU of 76.6%, a precision of 91.1%, and a recall of 83%.

Widiasri et al. [21] developed Dental-YOLO which is a modified YOLOv4 model for detecting AB and MC on 2D cross-sectional CBCT images. The model achieved mAP of 99.46%. The model also measured bone height and width using bounding box attributes. Statistical analysis showed no significant difference from manual measurements.

Widiasri et al. [22] proposed a U-Net-based model to segment the AB and MC from 2D cross-sectional CBCT images using two approaches: multiclass and binary segmentation. The model achieved a mIoU of 85% for simultaneous segmentation and 98% (AB) and 81% (MC) for separate binary segmentation. Naufal et al. [23] proposed a YOLOv8-based segmentation model for AB and MC on 2D cross sectional CBCT images and reconstructing them in 3D. The method aimed to support implant planning by providing accurate anatomical visualization. The proposed model achieved a DSC of 91.82%.

Although many deep learning approaches have been developed to identify the AB and MC, there are still several limitations. Most existing methods focus on detection or segmentation only and do not provide a direct recommendation of dental implant dimension. They still require clinicians to manually interpret the outputs. The previous study by Widiasri et al. [21] explored the use of a YOLO-based model, focusing on object detection to localize the anatomical regions of the AB and MC. However, the method was limited to coarse localization using bounding box attributes and did not perform pixel-wise segmentation, restricting its application to basic measurements of bone height and width. In addition, previous segmentation of AB and MC

approaches by Naufal et al. [23] using standard YOLOv8 architectures. It did not optimize computational efficiency or include any dental implant dimension recommendation. Furthermore, many deep learning methods in this area are based on computationally heavy architectures such as U-Net or 3D CNNs, limiting their practicality for real-time clinical use. Therefore, this study is motivated by the urgent requirement for a lightweight and accurate system that not only segments AB and MC but also automatically recommends dental implant dimensions which is not addressed by previous works.

You Only Look Once (YOLO) is a real-time object detection framework that has been developed across multiple versions, with continuous improvements in accuracy, speed, and architectural efficiency [24]. The YOLO series has developed from version 1 to the current version 12 [25]. Although it was initially developed for object detection tasks, later versions such as YOLOv8 can be used for segmentation tasks [26], [27]. It allows the model to perform both object detection and instance segmentation simultaneously. The flexibility and performance of YOLO have led to its growing adoption in various domains, including medical imaging [28].

The general architecture of YOLO consists of three main components: a backbone for extracting hierarchical features from the input image, a neck for aggregating and refining these features across different scales, and a head for producing the final outputs, such as bounding boxes, class labels, and segmentation masks [29]. YOLOv8 introduces several key improvements, including an anchor-free detection mechanism, a decoupled detection head for better task separation, and the use of efficient C2f modules within its backbone [30]. These enhancements made YOLOv8 as a robust and efficient solution for both detection and segmentation tasks, particularly in real-time medical image analysis [31]. However, applying the default YOLOv8 architecture to this task presents challenges due to the significantly different dimensions and characteristics of the AB and MC regions. The AB typically appears as a large structure while the MC is much smaller in cross sectional CBCT images. This discrepancy can lead to suboptimal segmentation performance, especially for the MC, as the multi-scale feature extraction in standard YOLOv8 does not sufficiently emphasize small structures within complex anatomical backgrounds.

This study proposes DIR-YOLO (Dental Implant Recommendation-YOLO), a modified version of YOLOv8n the lightweight n scale model specifically designed for the segmentation of AB and MC. In this model, the original C2f modules in the YOLOv8 backbone are replaced with C3Ghost modules, which leverage Ghost Convolution to reduce redundancy and computational load while preserving performance [32]. The standard convolution in YOLOv8 is also replaced with Ghost Convolution to reduce computational complexity. Additionally, the segmentation head is simplified to utilize only two-scale feature maps, focusing on high-scale features for segmenting small structures like the MC and low-scale features for larger structures like the AB.

These architectural modifications make DIR-YOLO more efficient and suitable for AB and MC segmentation. Beyond segmentation, the system also includes a module for recommending optimal dental implant dimensions, specifically diameter and height based on the segmented AB and MC.

II. METHODOLOGY

A. DATASET

This study used a dataset collected from the Universitas Airlangga Academic Dental Hospital, with ethical approval obtained from the Health Research Ethics Committee (certificate number: 00450/HRECC.FODM/IV/2024). All CBCT images were acquired using the OP300 3D scanner (Instrumentarium 161 Dental, Tuusula, Finland) with standardized imaging parameters. Fig. 2 illustrates the data collection process, including CBCT acquisition, cross-sectional CBCT image generation, anatomical annotation of the AB and MC, and expert selection of dental implant dimensions.

The initial dataset consisted of CBCT scans from 278 patients. After applying the inclusion criteria focused on edentulous molar regions, the dataset was refined to include 30 patients who had clearly identifiable edentulous areas in the posterior mandible suitable for dental implant planning. This subset resulted in a total of 607 cross-sectional CBCT images extracted from the selected patients. As illustrated in Fig.1, each edentulous region was processed to generate multiple cross-sectional CBCT images perpendicular to jaw curve, which serve as the standard views for dental implant planning. These cross-sectional images provide detailed visualization of both the AB and MC structures. It is important to note that within this dataset, no cases exhibited type 4 bone quality as defined by the Lekholm and Zarb classification [33], where the cortical bone is very thin and the trabecular bone is of low density. Therefore, none of the included patient represented the highest-risk bone condition described by Juodzbalsys et al. [34] for dental implant placement.

The use of 30 patients in this study aligns with the minimum sample size requirement based on the population proportion method described by Lemeshow et al. [35]. With an assumed prediction success rate of 90%, an absolute precision of 10%, and a 90% confidence level, the estimated minimum sample size needed is approximately 25 patients. Therefore, the inclusion of 30 patients not only satisfies but exceeds this threshold. It ensures that the evaluation of the proposed system is both statistically reliable and representative for assessing dental implant dimension recommendation performance.

The annotation of the AB and MC regions in each image was validated by two expert dental radiologists: the first dental radiologist (expert 1) and the second dental radiologist (expert 2) with 4 and 16 years of clinical experience in dental radiology and dental implant planning. This expert validation ensured the accuracy and clinical relevance of the ground truth segmentations. For each patient in the dataset,

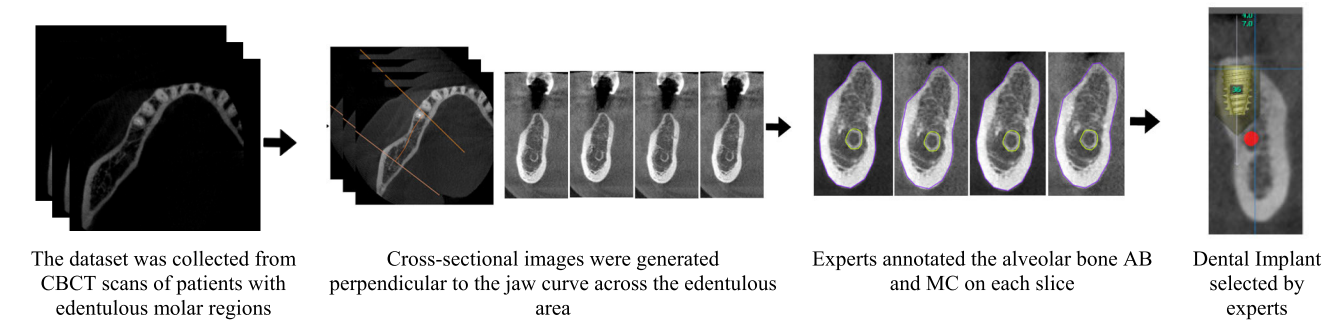


FIGURE 2. Data collection process.

TABLE 1. Intraclass Correlation Coefficient (ICC) between two expert radiologists for dental implant dimension recommendations (diameter and height).

		95% Confidence Interval			F Test with True Value 0			
		ICC	Lower Bound	Upper Bound	Value	df1	df2	Sig
Diameter	SM	0.93	0.86	0.97	28.21	29	29	<0.001
	AM	0.96	0.93	0.98	28.21	29	29	<0.001
Height	SM	0.99	0.97	0.99	146.52	29	29	<0.001
	AM	0.99	0.99	0.99	146.52	29	29	<0.001

SM: Single Measure, AM: Average Measure

appropriate dental implant dimensions were also determined independently by the same two experts. The expert independently assessed each edentulous region and recommended the optimal dental implant diameter and height based on standardized clinical protocols. Table 1 presents the Intraclass Correlation Coefficient (ICC) between the two experts. It demonstrates good agreement in their clinical assessments. This high level of concordance between them validates the reliability of the ground truth dental implant dimension recommendations used for training and evaluating our proposed method.

This study implemented a Leave-One-Out Cross-Validation (LOOCV) approach with 30 cross-validations to ensure robust assessment of the model's generalizability. In each of the 30 validation iterations, cross-sectional CBCT images from 29 patients were used as the training set while cross-sectional CBCT images from the 1 patient were used as testing set. This patient-based cross-validation strategy was specifically chosen to evaluate the model's performance on completely unseen patient data and mimicking real clinical scenarios.

B. DIR-YOLO

This study implemented DIR-YOLO, a computationally efficient modification of YOLOv8n specifically designed for

the segmentation of AB and MC. Fig. 3 shows the baseline architecture of YOLOv8n and Fig. 4 shows the architecture of DIR-YOLO. DIR-YOLO takes a 640×640 cross-sectional CBCT image as input. The backbone is responsible for extracting important features from the image. It uses GhostConv, which creates extra feature maps using simple and fast operations to reduce computation. It also uses C3Ghost modules, which combine ghost convolution with shortcut connections to make the feature extraction deeper and more efficient. The neck is designed based on Feature Pyramid Network (FPN) architecture which is widely used in object detection and segmentation tasks to combine low-level spatial features with high-level semantic features across different scales [36].

DIR-YOLO architecture implements three key modifications of YOLOv8 to enhance efficiency while maintaining segmentation performance. First, C2f modules are substituted with C3Ghost modules across both the backbone and neck of the network. C3Ghost is a lightweight convolutional module that integrates Ghost Convolution into the C3 module structure. It consists of multiple Ghost Bottlenecks embedded within a residual framework. The main advantage of C3Ghost lies in its ability to maintain rich feature representation while reducing computational cost and model complexity, which is especially beneficial for resource-constrained environments [37].

Second, standard convolution is replaced with Ghost Convolution (GhostConv) to reduce computational complexity. The GhostConv modules generate feature maps more efficiently by applying a small set of standard convolutions followed by cheaper linear operations to create additional "ghost" features [32].

Third, DIR-YOLO simplifies the segmentation head compared to standard YOLOv8. While YOLOv8 employs a complex three-scale design that combines low, medium, and high resolution features through multiple up-sampling and down-sampling operations, DIR-YOLO reduces this to only two feature scales. The low scale with 20×20 resolution focuses on segmenting large structures like AB, and the high scale with 80×80 resolution is used for small structures like MC. By removing the medium scale with 40×40 resolution, DIR-YOLO processes backbone features directly using

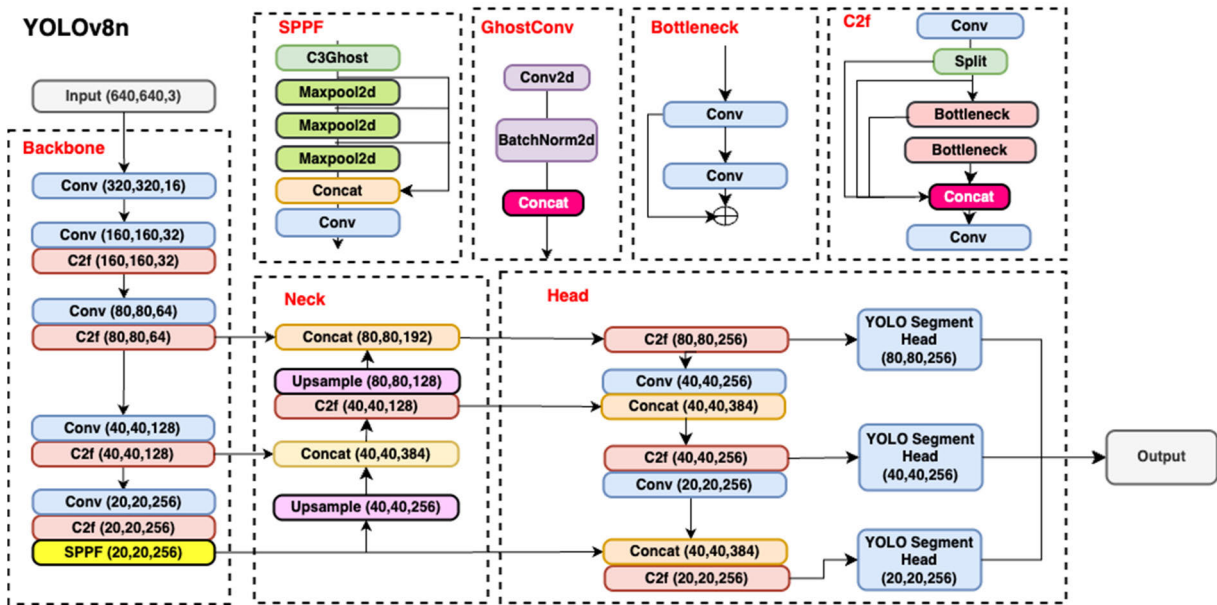


FIGURE 3. YOLOv8n baseline architecture.

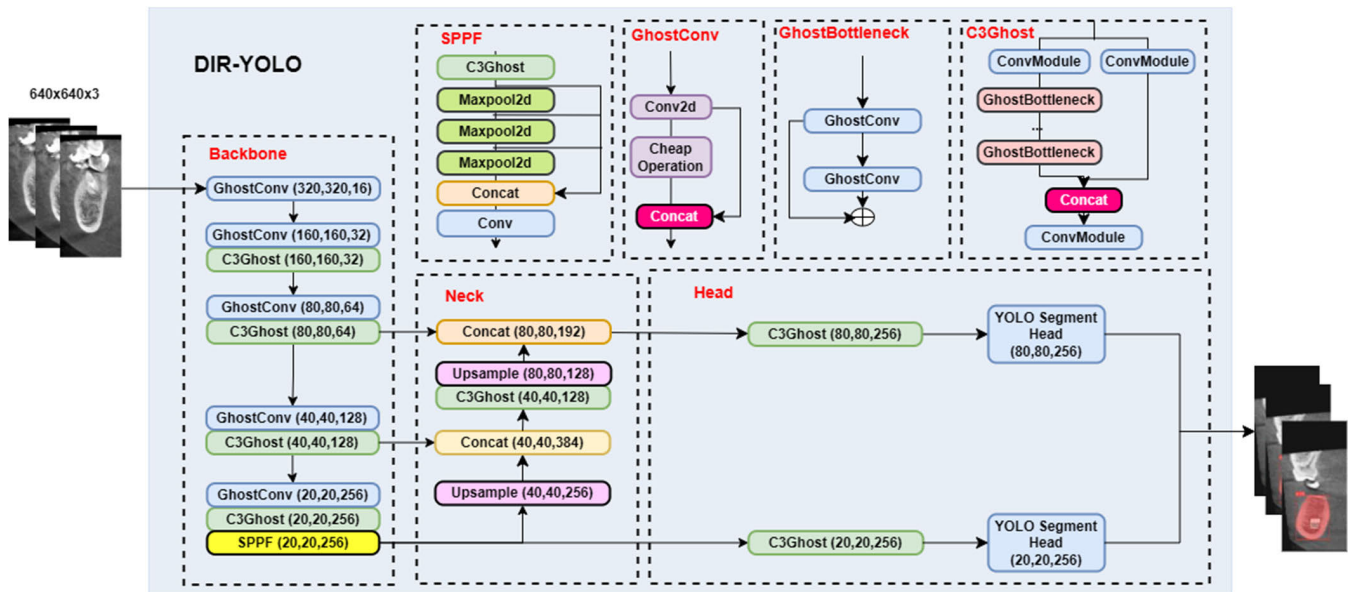


FIGURE 4. DIR-YOLO architecture.

a C3Ghost module and merges only the selected scales in the final segmentation layer. This design improves computational efficiency and maintaining segmentation performance.

DIR-YOLO specifically adopts the smallest-scale variant ('n' scale) from the YOLOv8 series to further reduce computational complexity. The 'n' scale is chosen due to its lower number of parameters and computational demands compared to larger-scale models ('s', 'm', 'l', or 'x'). By utilizing this lightweight configuration, DIR-YOLO effectively balances efficiency and accuracy, making it highly suitable for practical clinical deployment where computational resources may be limited.

C. DENTAL IMPLANT RECOMMENDATION FRAMEWORK

The proposed Dental Implant Recommendation Framework consists of several sequential steps designed to automate the dental implant selection process. Fig. 5 shows Dental Implant Recommendation Framework. First, the edentulous region is identified and selected in axial CBCT slices. From this region, cross-sectional CBCT images are generated following the natural curvature of the jaw arch. These slices serve as the primary input for DIR-YOLO segmentation model. The DIR-YOLO model then performs semantic segmentation on each slice for identifying the AB and MC. Based on these segmentation results, the system automatically determines

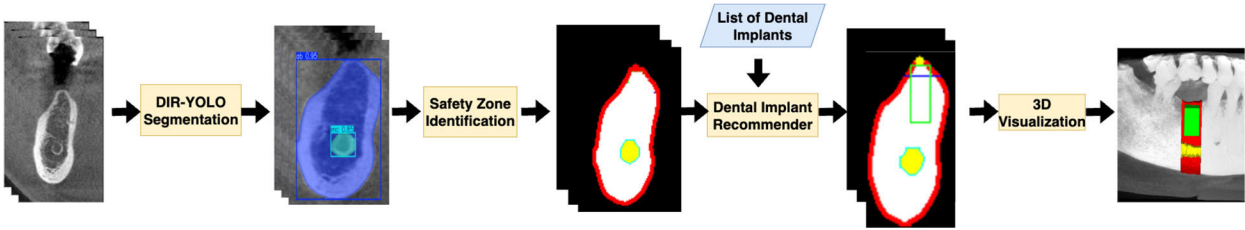


FIGURE 5. Dental implant recommendation framework.

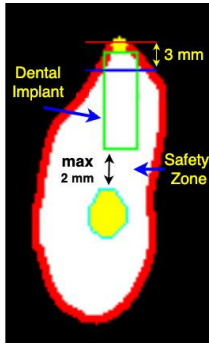


FIGURE 6. Dental implant is positioned within safety zone.

the safety zone which is a crucial region where dental implant placement is anatomically feasible. This study defines a safety zone as the region within the AB where a dental implant can be placed without risking damage to the MC or exceeding bone boundaries, following the guidelines proposed by Juodzbalys et al. [34].

In the CBCT images used, 1 pixel represents 0.3 mm in real-world measurements. The vertical safety margin is defined based on the distance from the MC. The dental implant must be placed at least 2 mm above the top of the MC to avoid nerve injury. The dental implant is placed starting from the center of the top of AB and the dental implant area that extends 3 mm from the top of the AB must be completely within the safety zone.

The horizontal safety margin ensures that the dental implant remains centered within the AB and maintains a distance of at least 1 mm from its outer edge. The horizontal safety margin is implemented using a morphological erosion technique applied to the segmented AB area. This image processing method reduces the outer boundary of the AB region to define a core area where dental implant placement is anatomically safe. The kernel used for erosion is a 3×3 pixel square. This dimension corresponds to a 1 mm margin, based on the image resolution where each pixel represents 0.3 mm. The erosion operation effectively shrinks the AB region inward by this distance. As a result, a more area referred to as the horizontal safety zone is created. This ensures that the dental implant is placed at least 1 mm away from the outer edge of the AB.

Fig. 6 presents an example of a cross-sectional CBCT image in which the dental implant is correctly positioned

TABLE 2. List of dental implant models with diameter and height.

Model Number	Diameter (mm)	Height (mm)	Model Number	Diameter (mm)	Height (mm)
GS2M3508O	3.5	8.5	GS2W5011O	5	11.5
GS2M3510O	3.5	10	GS2W5013O	5	13
GS2M3511O	3.5	11.5	GS2W5015O	5	15
GS2M3513O	3.5	13	MSP25103R	2	10
GS2M3515O	3.5	15	MSP25104R	2.5	10
GS2R4007O	4	7	MSP25113R	2.5	11.5
GS2R4008O	4	8.5	MSP25114R	2.5	11.5
GS2R4010O	4	10	MSP25133R	2.5	13
GS2R4011O	4	11.5	MSP25134R	2.5	13
GS2R4013O	4	13	MSP25153R	2.5	15
GS2R4015O	4	15	MSP25154R	2.5	15
GS2W4507O	4.5	7	MSP30103R	3	10
GS2W4508O	4.5	8.5	MSP30104R	3	10
GS2W4510O	4.5	10	MSP30113R	3	11.5
GS2W4511O	4.5	11.5	MSP30114R	3	11.5
GS2W4513O	4.5	13	MSP30133R	3	13
GS2W4515O	4.5	15	MSP30134R	3	13
GS2W5007O	5	7	MSP30153R	3	15
GS2W5008O	5	8.5	MSP30154R	3	15
GS2W5010O	5	10			

within the defined safety zone. The white region in the image represents the safety zone area, obtained by applying morphological erosion to the segmented AB to maintain a 1 mm margin from the outer bone boundary. While this visualization demonstrates a valid dental implant placement in a single slice, it is important to note that the implant must also fit entirely within the safety zone across all relevant cross-sectional CBCT images of edentulous region to ensure safe placement throughout the edentulous region. Based on the spatial dimensions of the safety zone, the system evaluates the available implant options and recommends the dimensions (diameter and height) that provide the greatest stability while complying with anatomical constraints.

The dental implant is selected from a predefined list, where each dental implant has a fixed diameter and height. This list forms a dental implant database, which can be updated or expanded according to the availability of implant products in clinical practice. In this study, the dental implant database was constructed based on commonly used implant dimensions at the Universitas Airlangga Academic Dental Hospital. Table 2

Algorithm 1 Pseudocode 1. Dental Implant Selection

```

Input:
  Slices    ← cross-sectional CBCT images
  AB_masks  ← segmented alveolar bone
  MC_masks  ← segmented mandibular canal
  Implants  ← list of implants (diameter,
height, volume)
  Resolution ← 0.3 mm per pixel
Output:
  RecommendedImplant
//Step 1: Define safety zone on each slice
For each slice in Slices:
  Apply erosion to AB_mask for horizontal
margin
  Mark region ≥ 2 mm above MC and ≥ 3 mm
inside AB as vertical safety zone
  Combine into full SafetyZone

//Step 2: Select baseline (middle) slices
BaselineSlices ← middle slice of edentulous
region

//Step 3: Filter implants that fit baseline
safety zone Sort Implants by volume
descending BaselineFit ← []
For implant in Implants:
  If implant fits in SafetyZone of
BaselineSlice:
    Add to BaselineFit

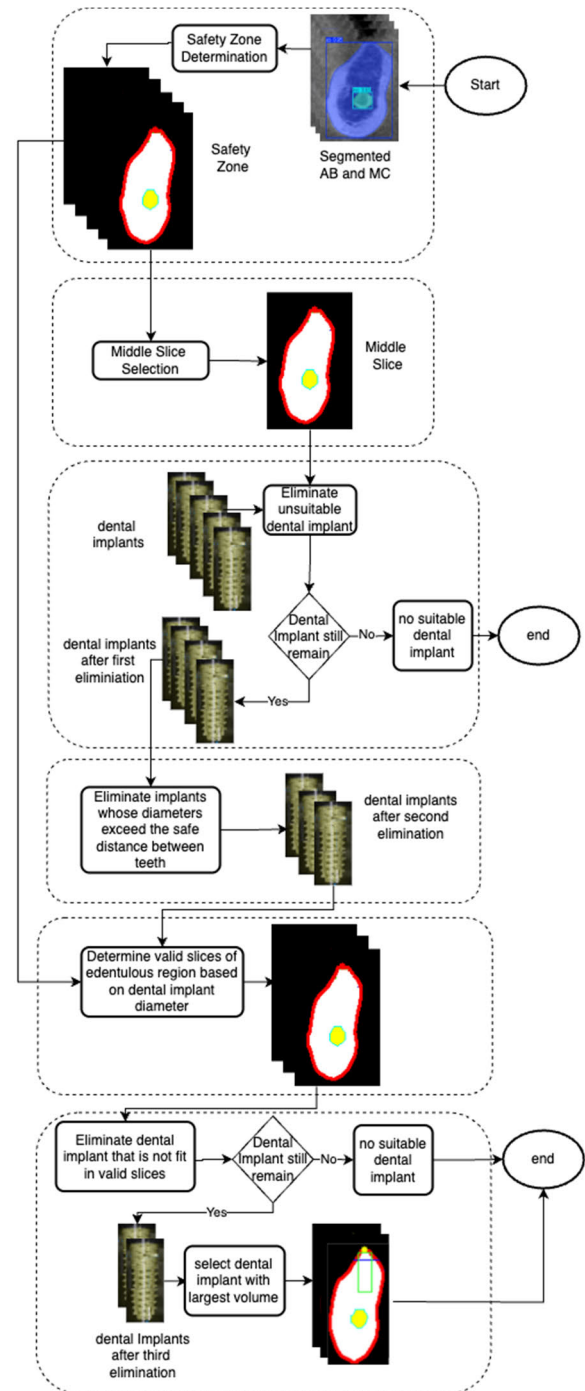
//Step 4: Filter by edentulous width
N ← total slices in edentulous region
We ← N × Resolution
Dmax ← We - 2 mm
DiameterFit ← []
For implant in BaselineFit:
  If implant.diameter ≤ Dmax:
    Add to DiameterFit

//Step 5: Final check on valid slices
(exclude first and last 3)
ValidSlices ← Slices excluding first and
last 3
FinalCandidates ← []
For implant in DiameterFit:
  If implant fits in SafetyZone of all
ValidSlices:
    Add to FinalCandidates

//Step 6: Select implant with largest volume
RecommendedImplant ← implant with max volume
in FinalCandidates

Return RecommendedImplant

```

**FIGURE 7.** Dental implant selection.

presents the complete list of available dental implant options used for evaluation and recommendation in this study.

The dental implant recommendation is performed through a multi-step evaluation using the segmented AB and MC across all cross-sectional CBCT images. Pseudocode 1 and Fig. 7 shows the steps involved in the dental implant selection process. First, the previously defined safety zone is applied to each AB and MC segmented area to identify the implantable

area. Among these slices, the middle slices of the edentulous region are chosen as the baseline slices, where the dental implant is initially tested for placement. A predefined list of dental implants, each with a fixed diameter and height, is sorted by volume in descending order. Each implant in the list is then virtually placed at the center of the middle slices. Dental implants that are not fully inside the safety zone in these slices are eliminated from the selection. If no implant meets the safety criteria at this stage, the system will recommend that no suitable implant is available

The remaining dental implant candidates that fit within the middle slices proceed to the next filtering step, which considers the total width of the edentulous region to determine the allowable dental implant diameter. Since each cross-sectional images represents 0.3 mm in real measurement, the total edentulous width W_e is calculated using the formula given by (1), where N is the number of slices in the edentulous region.

$$W_e = N \times 0.3 \quad (1)$$

To ensure anatomical safety, the dental implant must maintain a 1 mm distance from each neighbouring tooth, resulting in a 2 mm safety margin. Therefore, the maximum allowable implant diameter $D_{\max}$ is given by (2).

$$D_{\max} = W_e - 2 \quad (2)$$

Dental implant candidates with a diameter greater than $D_{\max}$ are eliminated from the selection. For example, if the edentulous region contains 17 slices, the total width is $W_e = 17 \times 0.3 = 5.1$ mm and the maximum dental implant diameter allowed is $D_{\max} = 5.1 - 2 = 3.1$ mm.

The final step involves testing the remaining group of dental implants across all valid slices within the edentulous region. To ensure safety based on the width of the edentulous region, the first and last three slices of the region, which correspond to approximately 2 mm, are excluded from implant placement because these boundary areas fall outside the defined safety zone. The remaining slices are used to evaluate the candidate dental implants, which are virtually placed on each of these slices. Only dental implants that fully fit within the safety zone on every tested slice are considered suitable. From this final filtered list, the dental implant with the largest volume is selected as the recommended dental implant.

The final output provides a 3D visualization that displays the recommended dental implant placed within the segmented anatomical structures. It includes the AB and MC both of which are visualized based on the segmentation results. This combined visualization allows clinicians to assess the dental implant's position in relation to the AB and MC.

D. EXPERIMENTAL DESIGN

The proposed DIR-YOLO model was trained and evaluated using a dataset consisting of CBCT scans from 30 patients with edentulous molar regions. To ensure reliable model assessment, this study applied a Leave-One-Out Cross-Validation (LOOCV) approach. LOOCV is a well-established strategy in medical imaging research and allows for detailed performance evaluation across individualized patient anatomy [38]. In LOOCV, the model is trained on data from 29 patients and tested on the remaining one, repeating this process 30 times so that each patient serves as the test case once. This approach ensures that the model is evaluated on unseen patient data in every iteration. During each LOOCV fold, the number of training images from the 29 patients varies depending on the number of cross-sectional

slices extracted from the edentulous regions. As a result, the training set typically contains between 570 and 600 images per fold. Likewise, the number of images in the testing set, which corresponds to the remaining patient, ranges from approximately 10 to 30 images depending on the dimension of the edentulous region for that patient. This setup reflects a realistic variation in clinical imaging and supports a comprehensive evaluation at the image and patient level. To enhance model performance and prevent overfitting, the training dataset was augmented using Roboflow which effectively tripled the dataset dimension by generating three versions of each image (the original plus two augmented variants). With this augmentation, the total number of training images per fold ranges from approximately 1,710 to 1,800. This augmentation process incorporated multiple transformations including horizontal flips, rotations, cropping, shear transformations, hue adjustments, saturation variations, brightness modifications, exposure changes, blur effects, and noise addition. Data augmentation was performed only on slices from the training set and not on the test patient, thereby preventing data leakage from correlated slices. Additionally, 10% of the training set was randomly selected to serve as a validation set to monitor the performance of the model during each training epoch. The model demonstrating the highest performance on the validation set in specific epoch was then selected as the final model for evaluation on the testing set.

The training process was implemented using the Ultralytics framework with default training parameters, including a batch dimension of 16, SGD optimizer with an initial learning rate of 0.01, and image dimension of 640×640 pixels. The training was run for 100 epochs in each fold of the cross-validation to ensure convergence and stable performance.

The segmentation performance of DIR-YOLO for the AB and MC was evaluated using Dice Similarity Coefficient (DSC). DSC was chosen as the primary evaluation metric due to its greater sensitivity to overlap. It especially suitable for medical image segmentation where precise anatomical boundaries are critical [39]. The formula for calculating the DSC for each structure $X \in \{AB, MC\}$, based on the overlap between the predicted mask P_X and ground truth mask G_X is given by (3). The overall DSC per patient is the average of the AB and MC DSC values. The mean DSC (mDSC), also given by (4), represents the average DSC across all N patients in the dataset. Each DSC_i corresponds to the overall DSC computed for the i -th patient.

$$DSC_X = \frac{2 \times |P_X \cap G_X|}{|P_X| + |G_X|}, X \in \{AB, MC\}$$

$$DSC = \frac{1}{2} (DSC_{AB} + DSC_{MC}) \quad (3)$$

$$mDSC = \frac{1}{N} \sum_{i=1}^N DSC_i \quad (4)$$

The boundary accuracy of DIR-YOLO for AB and MC was further evaluated using the 95th percentile Hausdorff Distance (HD95) [40]. While DSC measures the degree of

overlap between predicted and ground truth regions, it can be insensitive to small but clinically important boundary errors, particularly for thin structures such as the mandibular canal. The Hausdorff Distance (HD) quantifies the maximum deviation between two boundaries, defined as the maximum of the minimum distances between points on one boundary to the other. However, it is highly sensitive to outliers, such as single mislabeled pixels, annotation noise, or irregularities in the ground truth boundaries of AB and MC. In such cases, the maximum distance may be dominated by one extreme deviation, giving a misleading impression of large segmentation errors even when the rest of the boundary is closely aligned. To mitigate this limitation, HD95 [40] was chosen in this study, as it represents the 95th percentile of all boundary distances rather than the absolute maximum. This makes HD95 a more robust and stable metric for medical image segmentation, reducing the influence of isolated annotation errors or noise while still capturing clinically meaningful deviations. In practice, HD95 ensures that the reported error reflects the typical boundary precision of the segmentation, rather than being distorted by a small number of extreme outliers.

The HD95 between predicted P_X and ground truth G_X , for each structure $X \in \{AB, MC\}$, is defined by (5).

$$\begin{aligned} & \text{HD95}_X(P_X, G_X) \\ &= \max \text{ percentile95 } \left\{ \sup_{x \in P_X} \inf_{y \in G_X} d(x, y), \sup_{y \in G_X} \inf_{x \in P_X} d(x, y) \right\} \end{aligned} \quad (5)$$

The overall mean HD95 per patient was then calculated as the average of the AB and MC HD95 values by (6). Finally, the mean HD95 (mHD95) across all N patients, representing the average boundary deviation across the dataset, was obtained by (7).

$$\text{HD95} = \frac{1}{2} (\text{HD95}_{AB} + \text{HD95}_{MC}) \quad (6)$$

$$\text{mHD95} = \frac{1}{N} \sum_{i=1}^N \text{HD95}_i \quad (7)$$

In addition, the computational efficiency of the model was also evaluated. Two commonly used indicators were measured: the number of parameters, which reflects the model dimension, and GFLOPs (Giga Floating Point Operations per second), which represents the computational cost during inference. These metrics are essential for assessing the model's suitability for real-time and resource-constrained clinical applications [41].

The evaluation of the dental implant recommendation involved two expert dental radiologists (expert 1 and expert 2), who independently selected appropriate dental implant dimensions (diameter and height) for each edentulous region using the OnDemand3D Dental software. The agreement and consistency of implant selection between the

experts and the proposed method were analyzed using a series of statistical validation techniques. These included Cohen's Weighted Kappa to evaluate the agreement between the system and expert recommendations [42], as it effectively measures consistency for ordinal data like dental implant dimensions. The Kruskal–Wallis test was used to assess whether there were statistically significant differences in implant dimensions among Expert 1, Expert 2, and the system. It is suitable for comparing non-normally distributed data across multiple independent groups [43]. Additionally, the Mean Absolute Error (MAE) was used to quantify the average prediction error in implant dimensions. This combination of statistical measures ensures that the evaluation is both quantitative and statistically validated and provides strong evidence of the model's effectiveness and clinical relevance.

III. RESULTS AND DISCUSSION

A. SEGMENTATION PERFORMANCE

The segmentation performance of DIR-YOLO model was evaluated by comparing it with other YOLO-based segmentation models. The YOLOv8 segmentation model was used as the baseline model to assess the improvements introduced by DIR-YOLO. Additionally, the model was compared with the latest versions of YOLO, including YOLOv9, YOLOv10, YOLOv11, and YOLOv12, to evaluate its effectiveness against newer variants in the same family. For a fair and efficient comparison, all YOLO models from versions 8 to 12 were implemented using the 'n' scale variant, while YOLOv9 used the 't' (tiny) variant. These configurations represent the smallest and most lightweight versions of each model.

The segmentation results were also evaluated against previous studies in the same domain that focused on segmenting the AB and MC from cross-sectional CBCT images. Widiastri et al. [21] applied a U-Net based architecture, while Naufal et al. [23] implemented the YOLOv8 medium-scale model (YOLOv8m) for the same task. These comparisons help show how well DIR-YOLO performs compared to other existing studies.

Table 3 shows the segmentation results of the AB and MC across several cross sectional CBCT images. The green and cyan contours represent the ground truth segmentations of the AB and MC, respectively, while the red and blue contours indicate the predicted segmentations produced by each model. Table 4 presents the segmentation performance comparison between the proposed DIR-YOLO model and various existing segmentation approaches. This table serves not only as a performance benchmark but also as an ablation study with respect to the proposed DIR-YOLO modifications. By comparing DIR-YOLO (which integrates C3Ghost, Ghost Convolution, and two-scale head simplification) with existing YOLO variants that do not use these modules, the proposed architectural modifications can be observed.

TABLE 3. Visual comparison of AB and MC segmentation results from various models.

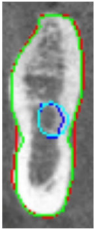
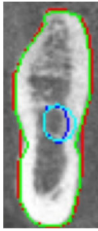
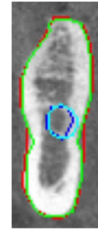
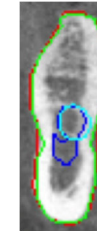
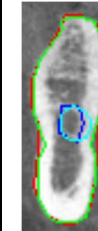
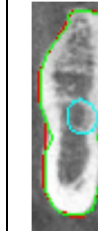
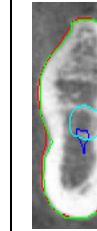
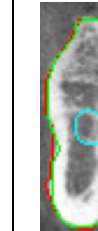
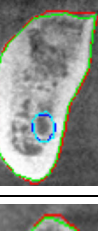
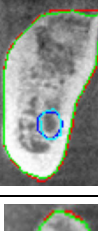
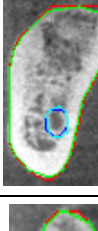
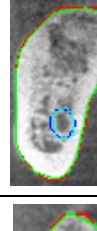
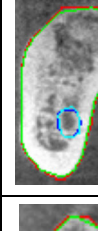
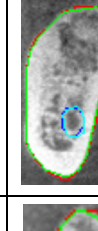
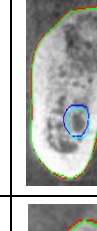
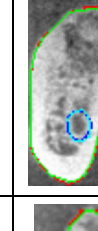
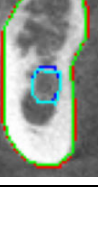
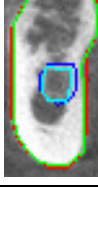
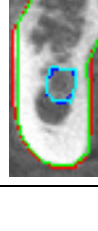
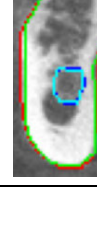
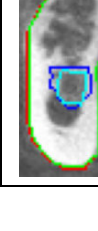
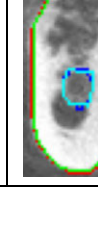
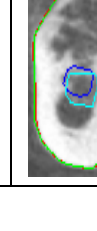
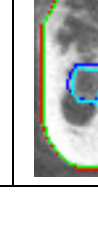
Image	DIR-YOLO (ours)	YOLOv8n	YOLOv9t	YOLOv10n	YOLOv11n	YOLOv12n	U-NET [21]	YOLOv8m [22]
1								
2								
3								

TABLE 4. Segmentation performance comparison (ablation with existing YOLO models).

Model	DSC AB (%)	DSC MC (%)	mDSC (%)	HD95 AB (mm)	HD95 MC (mm)	mHD95 (mm)	Num of params	GFLOPS
DIR-YOLO (ours)	97.65	86.59	92.12	0.5241	0.7304	0.6273	1,521,856	7.8
Yolov8n	97.65	83.77	90.71	0.5061	0.9742	0.7982	3,263,990	12.1
Yolov9t	97.66	84.93	91.29	0.5775	0.8325	0.6768	3,093,542	54.1
Yolov10n	97.57	85.58	91.58	0.7507	0.9775	0.9424	2,529,542	10.6
Yolov11n	97.59	83.65	90.62	0.8657	0.8940	0.7001	2,842,998	10.4
Yolov12n	97.69	83.32	90.5	0.5994	0.8386	0.7080	2,821,206	10.4
U-Net [21]	95.72	70.75	83.24	1.5547	0.8438	1.8990	31,040,000	41.04
Yolov8m [22]	97.58	82.48	90.03	0.6477	0.9276	0.8967	27,240,790	110.4

DIR-YOLO achieves the highest mDSC at 92.12%, with the best DSC for MC segmentation (86.59%) while maintaining a competitive DSC for AB (97.65%). Although YOLOv12n slightly outperforms DIR-YOLO in AB segmentation (97.69%), its performance on MC is lower (83.32%), resulting in a lower overall mDSC (90.50%). In terms of boundary accuracy, DIR-YOLO achieves the lowest mean Hausdorff Distance 95 (mHD95: 0.6273 mm), clearly outperforming other YOLO variants (0.6768–0.9424 mm) and prior studies such as U-Net [22] (1.8990 mm) and YOLOv8m [23] (0.8967 mm). When considering individual structures, DIR-YOLO attains HD95 AB = 0.5241 mm and HD95 MC = 0.7304 mm, which correspond to boundary deviations of approximately 2–3 pixels in CBCT (0.3 mm/pixel). These

deviations remain well within the recommended 1 mm horizontal and 2 mm vertical safety margins for implant planning by Juodzbalsys et al. [34]. This is particularly important for the MC, where DIR-YOLO records the lowest HD95 among all tested models, ensuring safer dental implant placement by minimizing the risk of nerve injury due to segmentation errors. By contrast, U-Net shows much higher errors (HD95 AB = 1.5547 mm; HD95 MC = 0.8438 mm; mHD95 = 1.8990 mm), confirming its limited ability to capture fine anatomical structures. Moreover, DIR-YOLO demonstrates exceptional computational efficiency, with the lowest number of parameters (1.52 million) and lowest GFLOPS (7.8) among all compared models. In contrast, U-Net and YOLOv8m exhibit substantially higher complexity, with over

TABLE 5. Ablation study of DIR-YOLO: comparison of head scale configurations and Ghost module contributions for AB and MC segmentation.

Model	DSC AB (%)	DSC MC (%)	mDSC (%)	HD95 AB (mm)	HD95 MC (mm)	mHD95 (mm)	Num of params	GFLOPS
Head configuration ablation (with C3Ghost + GhostConv applied):								
low-high (ours)	97.65	86.59	92.12	0.5241	0.7304	0.6273	1,521,856	7.8
low-medium-high	97.65	83.77	90.71	0.5061	0.9742	0.7982	3,263,990	12.1
low-medium	97.66	84.93	91.29	0.5775	0.8325	0.6768	3,093,542	54.1
medium-high	97.57	85.58	91.58	0.7507	0.9775	0.9424	2,529,542	10.6
Ghost module ablation (all using low-high head):								
No modification	97.59	83.65	90.62	0.8657	0.8940	0.7001	2,649,620	10.4
C3ghost only	97.69	83.32	90.5	0.5994	0.8386	0.7080	1,716,072	8.3
GhostConv only	95.72	70.75	83.24	1.5547	0.8438	1.8990	2,459,764	10.1

27 million parameters and GFLOPS exceeding 40 and 100, respectively. These results highlight that DIR-YOLO not only balances segmentation accuracy and computational efficiency but also provides superior boundary precision, which is clinically critical for safe dental implant planning.

An ablation study was conducted to evaluate both the effect of segmentation head scale configurations and the contribution of Ghost modules in DIR-YOLO and it is shown in Table 5. For the head configuration analysis, four variations were tested: low-high (proposed), low-medium-high, low-medium, and medium-high. The low-high configuration consistently outperforms the other variants. It achieves the highest overall mDSC (92.12%), the best MC segmentation mDSC (86.59%), and a competitive AB mDSC (97.65%). By comparison, configurations that include the medium scale (low-medium-high, low-medium, and medium-high) show a decline in MC segmentation performance, with mDSC values dropping to 83.64–85.35%. In terms of boundary accuracy measured by HD95, the low-high configuration also demonstrates the most reliable performance, with the lowest HD95 AB (0.52 mm) and HD95 MC (0.73 mm). This indicates that removing the medium scale not only improves segmentation accuracy but also reduces computational complexity. Based on these findings, the low-high head configuration was selected as the backbone for further Ghost module ablation. Three versions were evaluated: no modification (baseline YOLOv8n), C3Ghost only, and GhostConv only. The results show that C3Ghost alone improves AB segmentation (97.69%) but lowers MC accuracy, while GhostConv alone reduces both mDSC and boundary accuracy, highlighting its limited effectiveness when applied in isolation. In contrast, the combination of C3Ghost and GhostConv within DIR-YOLO provides the best balance between accuracy and efficiency. Using the low-high head scale for this comparison was necessary because it had already been shown to be the strongest performer among head configurations, ensuring that the evaluation of Ghost modules was conducted under the most effective head design.

FPN-based architectures employ three scales. However, in dental CBCT images, the most relevant information comes from the smallest structures (MC) and the largest structures (AB). The medium scale did not provide additional useful information and instead reduced segmentation performance. For this reason, the FPN design was adapted to two scales, which better suits the anatomical characteristics of this task.

The enhanced performance of DIR-YOLO can be attributed to a set of architectural modifications applied to the original YOLOv8n framework. These include the replacement of the standard C2f module with the more efficient C3Ghost module, the substitution of regular convolution layers with Ghost Convolution, and a refinement of the segmentation head to operate on low and high resolution feature scales only. These changes significantly enhance the model's ability to segment both the AB and MC. C3Ghost improves feature representation by integrating lightweight ghost bottlenecks with residual connections. It enables the network to capture richer semantic features with fewer parameters. Ghost Convolution generates more feature maps using cheaper linear operations. It reduces computational redundancy while preserving essential spatial information. Additionally, reducing the head to only low and high scales helps the model focus on large structures like AB and small detailed structures like MC. It minimizes unnecessary processing from the mid-scale path. Together, these modifications improve DIR-YOLO's efficiency and accuracy, especially in handling complex anatomical structures such as the MC, without increasing computational cost. This architectural optimization explains DIR-YOLO's superior segmentation performance and computational efficiency compared to the original YOLOv8 and other benchmark models.

B. DENTAL IMPLANT RECOMMENDATION PERFORMANCE

The segmentation results of the AB and MC from DIR-YOLO serve as the basis for recommending the appropriate dental

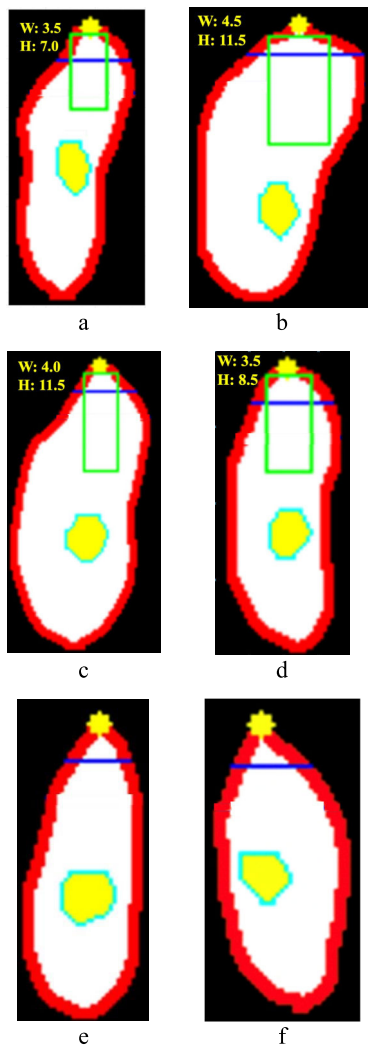


FIGURE 8. Results of dental implant recommendations by the system on edentulous regions. (a–d) show cases where the system recommends dental implant placement. In (e–f), the system does not recommend any implant due to the sharply narrow bone crest.

implant dimensions. Fig. 8 shows the dental implant recommendation results on several middle slices of cross sectional CBCT images in edentulous region. In each image, a green rectangular box represents the virtually placed implant, with its width corresponding to the implant diameter and its height to the vertical length. The blue horizontal line indicates the boundary located 3 mm below the crest of the AB. Images (a) through (d) show examples where the system identifies sufficient anatomical space for safe dental implant placement based on the available implant dimensions and the defined safety zone. These regions meet the required horizontal and vertical margins from the AB boundaries and the MC. In contrast, images (e) and (f) illustrate cases where the crest of the AB is too sharply pointed, and as a result, the system finds no suitable dental implant from the available database that fits within the defined safety zone. Consequently, the system does not recommend any implant for these slices. These conditions indicate that pre-treatment, such as bone augmentation or

TABLE 6. Results of weighted Cohen's kappa.

Ratings	Weighted Kappa	Asymptotic			95% Asymptotic Confidence Interval	
		Std. Error	z	Sig. (p)	Lower Bound	Upper Bound
Diameter expert 1 - System	0.835	0.084	4.599	<0.001	0.67	1.001
Diameter expert 2 - System	0.84	0.084	4.649	<0.001	0.675	1.005
Height expert 1 - System	0.93	0.045	5.141	<0.001	0.843	1.018
Height expert 2 - System	0.929	0.046	5.11	<0.001	0.839	1.019

TABLE 7. MAE and SD of dental implant dimensions between the system and two experts.

	MAE (SD)	
	diameter	height
System - Expert 1	0.25 (0.39)	0.25 (0.69)
System - Expert 2	0.23 (0.39)	0.25 (0.69)

TABLE 8. Kruskal-Wallis test results comparing dental implant size recommendations from two experts and the system.

	diameter	height
Kruskal-Wallis H	0.742	0.256
df	2	2
Asymp. Sig. (p value)	0.69	0.88

reshaping, is required before implant placement can be safely considered.

Table 6 presents the Weighted Cohen's Kappa results for agreement between the system and two experts, evaluated separately for dental implant diameter and height. The Weighted Kappa values range from 0.835 to 0.93, indicating a strong to almost perfect agreement. The high values, along with statistically significant p -values ($p < 0.001$), confirm that the system's recommendations are highly consistent with expert clinical judgment.

Table 7 presents the MAE and SD between the system's recommended dental implant dimensions and those provided by expert 1 and expert 2. The MAE for both diameter and height is low, with values ranging from 0.23 mm to 0.25 mm, indicating high accuracy of the system's predictions. The standard deviations, 0.39 mm for diameter and 0.69 mm for height, reflect consistent agreement across cases. These results demonstrate that the system's recommendations closely align with expert with minimal deviation.

Table 8 presents the results of the Kruskal-Wallis test used to assess whether there are statistically significant differences

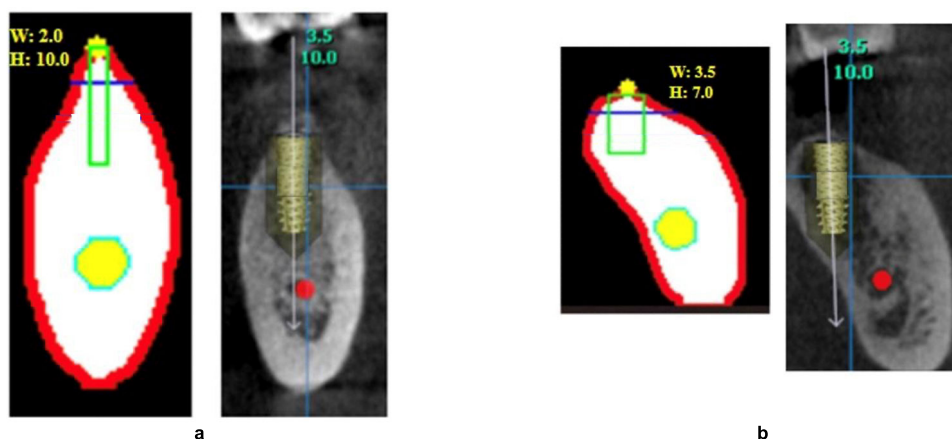


FIGURE 9. Discrepancy between the system's prediction (left) and expert recommendation (right) for dental implant placement. Subfigure (a) shows a discrepancy in diameter, while subfigure (b) highlights a discrepancy in height.

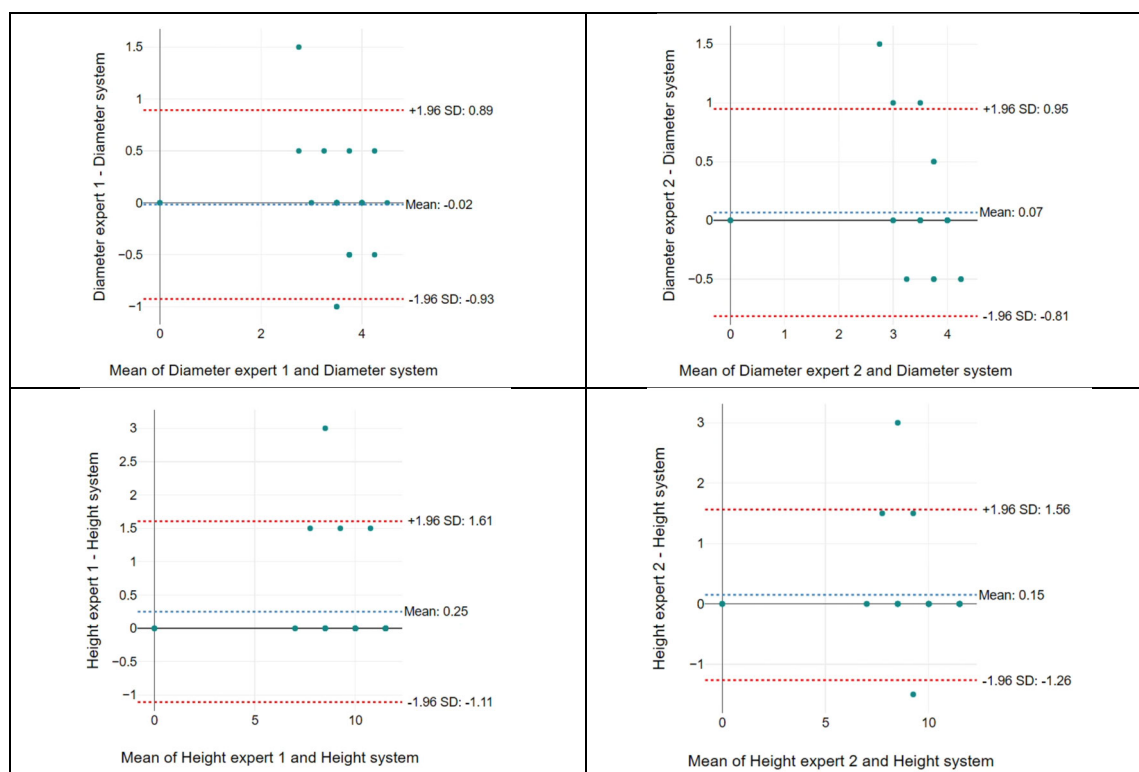


FIGURE 10. Bland-Altman plots comparing dental implant diameter and height recommendations between the system and two dental radiologists (DR 1 and DR 2).

in the dental implant diameter and height recommendations made by expert 1, expert 2, and the system. For diameter, the kruskal-Wallis H value is 0.742 with a p-value of 0.69, and for height, the h value is 0.256 with a p-value of 0.88. Both p-values are well above the standard significance threshold ($p > 0.05$), indicating that there is no statistically significant difference among the three groups. This result supports the conclusion that the system's dental implant recommendations are comparable to those made by experts.

A notable discrepancy between the system's prediction and expert recommendations was observed as illustrated in Fig. III-B(a). In this case, the system suggested a smaller implant diameter than the experts. This occurred because the patient's AB had a sharp crest, which limited the available width near the top. Although the system correctly identified a safe implant option, the expert chose to lower the implant placement point by approximately 3 mm to a wider and more stable region of the bone. This adjustment ensured better implant stability and safety. The visual example highlights

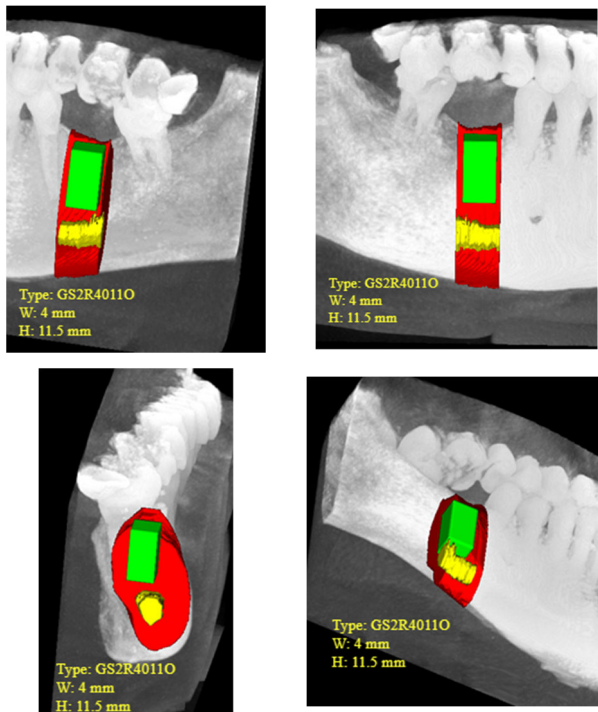


FIGURE 11. Visualization of the 3D reconstructed model from segmented AB and MC with the recommended dental shown from multiple viewing angles through model rotation.

the importance of additional clinical and visual assessment in anatomically challenging cases, even when the system's prediction aligns with the defined safety zone.

In another case, a discrepancy was observed in the predicted implant height. As shown in Fig. III-B(b), the AB appeared relatively regular in shape but was limited in height near the crest, which led the system to recommend a shorter implant (3.5 mm × 7 mm) based on the predefined safety zone. The system conservatively calculates the vertical safety zone by maintaining at least 2 mm of clearance above the MC and avoiding placement too close to the AB crest. However, the expert selected a longer implant (3.5 mm × 10 mm) by adjusting the implant position slightly deeper within the bone. They target a region with greater vertical bone support for improved stability. This decision was made based on anatomical judgment and visual assessment. This case shows that although the system follows safety rules, expert evaluation is still important for achieving the most stable and suitable implant placement.

Fig. 10 shows Bland–Altman plots comparing the system's implant diameter and height recommendations with those of two experts. The plots indicate a small mean difference and most points lie within the 95% limits of agreement, suggesting good consistency between the system and expert evaluations.

C. VISUALIZATION OF DENTAL IMPLANT

The visualization of dental implant was conducted through a 3D reconstruction approach. This process involves stacking

the segmented results of the AB, MC, and dental implant across all cross-sectional CBCT images to form a comprehensive 3D representation of the patient's jaw. To improve visual continuity between slices, interpolation was applied for ensuring a smoother and more realistic appearance of the AB, MC, and dental implant in the reconstructed volume. A blending technique was used to merge all three objects into a single visual model, where the AB was rendered with a translucent no-depth effect to avoid obscuring the view of the MC and dental implant. Meanwhile, the MC and dental implant were visualized with translucent to maintain clear spatial relationships. Rendering was performed using isosurface rendering to focus on the main objects (AB, MC, and dental implant) and Maximum Intensity Projection (MIP) to provide a global context of the entire CBCT data.

The reconstructed model was visualized using Napari, an interactive 3D visualization tool that allows users to rotate, zoom, and explore the model intuitively. Fig. 11 shows an example of the 3D reconstructed model from segmented AB and MC with the recommended dental implant placed in a patient jaw. It shows that the best recommended implant for this case is GS2R4011O, which has a diameter of 4.0 mm and a height of 11.5 mm. This dental implant was selected by the system based on its optimal fit within the defined safety zone across multiple slices. This visualization enables clinicians to evaluate the dental implant's position relative to surrounding anatomical structures. However, a limitation of this 3D visualization is that the dental implant is currently represented in a cuboid form rather than a cylindrical shape. This simplification was made to match the voxel-based format of the CBCT segmentation data, which uses uniform 2D pixel slices. As a result, a precise cylindric representation of the dental implant was not feasible within the current implementation.

IV. CONCLUSION AND FUTURE WORKS

Research on automated dental implant prediction remains limited, and there is a strong need for systems that can support clinicians in making accurate and safe decisions. This study proposed DIR-YOLO, a lightweight and efficient segmentation model based on a modified YOLOv8 architecture, specifically designed for accurate segmentation of the AB and MC in cross-sectional CBCT images. The model introduces three key architectural changes: the replacement of the C2f module with C3Ghost, the substitution of standard convolutions with Ghost Convolution, and the simplification of the YOLOv8 segmentation head to utilize only two feature scales. These modifications enhance feature extraction and reduce computational complexity while maintaining high segmentation accuracy. As a result, DIR-YOLO achieves superior performance in segmenting both AB and MC, particularly excelling in MC segmentation, and outperforms existing YOLO variants and traditional models like U-Net in terms of mDSC, mHD95, and computational efficiency.

DIR-YOLO incorporates an automated framework for dental implant dimension recommendation by utilizing the segmented AB and MC, the calculated safety zone, and a predefined database of available dental implants. The system selects the optimal dental implant that fit within the anatomical constraints. Statistical evaluations, including Weighted Cohen's Kappa, MAE, and the Kruskal-Wallis test, demonstrate strong agreement between the system's recommendations and expert selections, confirming the clinical reliability of the proposed method.

DIR-YOLO shows promising results, but this study still has some limitations. The current system can only recommend implants for single edentulous regions, so it does not yet handle cases with multiple adjacent missing teeth. The 3D visualization of implants is also simplified into cuboid shapes rather than their true cylindrical form. This simplification has clinical implications, because the cuboid corners extend beyond the true circular boundary of a cylindrical dental implant, the system may sometimes reject a dental implant that would clinically fit within the available bone. This leads to a conservative bias (false negatives), which reduces the number of accepted implants but ensures safety by not approving implants that could intersect critical structures.

Future work will begin by extending the system to address more complex clinical cases. The current framework only supports implant recommendations for single edentulous regions, which limits its applicability. Expanding its capability to handle multiple adjacent missing teeth, including multi-rooted regions, will make the system more versatile and clinically comprehensive.

In addition to expanding case complexity, future work should also incorporate bone quality analysis into the prediction framework. Since bone quality is determined by the combination of cortical thickness and trabecular density, its inclusion is critical for evaluating primary stability and long-term implant success. By developing an automated classification model of bone quality prior to implant prediction, the system could provide not only implant dimensions but also risk factor assessments. This would connect geometric feasibility with biological considerations and enhance the clinical relevance of the framework.

Another important direction for improvement is 3D visualization. At present, implants are represented using cuboid shapes, which introduces geometric distortion and may result in conservative recommendations. Transitioning from cuboid to cylindrical rendering will provide a more realistic representation of implant geometry and reduce the likelihood of rejecting implants that would otherwise be clinically acceptable. This refinement will also ensure that the system's visualization is better aligned with clinical practice.

Future work will also explore super-resolution techniques to overcome the limitations of fixed input resolution

(640×640) in CBCT images. With each pixel corresponding to approximately 0.3 mm, even small errors can significantly affect the delineation of the MC, which is critical for preventing nerve injury. Recent advances in hybrid deep learning methods, such as the GAN-Transformer framework introduced by Asif et al. [44], have shown strong potential for enhancing image resolution and recovering fine structural details. Applying such approaches to CBCT may improve the clarity of delicate anatomical structures, including both the AB and MC. In addition, future research will explore combining handcrafted features with deep learning representations to further improve boundary precision. Ullah et al. [45] demonstrated that fusing handcrafted descriptors (e.g., ELBP, DWT, GLCM) with CNN-based pathways significantly improved segmentation of fine tumor boundaries in brain MRI. Adapting this type of hybrid strategy to dental CBCT could complement the domain-specific two-scale head design in DIR-YOLO and yield more accurate delineation of AB and MC boundaries than purely data-driven approaches alone.

Finally, integration into a real-time clinical environment will be essential to ensure practical usability. This will require optimizing inference efficiency, developing intuitive visualization tools, and ensuring compatibility with dental imaging workflows. Achieving these steps will move the system closer to becoming a fully automated and clinically viable AI-assisted solution for dental implant planning.

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